

Exercises 3.2

1. Let I be the given set, which is certainly non-empty and contained in $M_2(\mathbb{R})$.

Clearly $A - B \in I$ for all $A, B \in I$.

Also $\begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix} \begin{pmatrix} p & q \\ r & s \end{pmatrix} = \begin{pmatrix} ap + br & aq + bs \\ 0 & 0 \end{pmatrix} \in I$, so I is closed under post-multiplication by elements of $M_2(\mathbb{R})$. Thus I is a right ideal of $M_2(\mathbb{R})$.

$\begin{pmatrix} p & q \\ r & s \end{pmatrix} \begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} pa & pb \\ ra & rb \end{pmatrix} \notin I$ in general, so I is not a left ideal of $M_2(\mathbb{R})$.

2. Clearly Rx is a non-empty subset of R .

Let $a, b \in Rx$, so $a = yx$ and $b = zx$ for some $y, z \in R$.

Then $a - b = yx - zx = (y - z)x$. As R is a ring, $y - z \in R$, so $a - b \in Rx$.

Also, for any $r \in R$, $ra = r(yx) = (ry)x$. As R is a ring, $ry \in R$, so $ra \in Rx$.

Thus Rx is a left ideal of R .

3. By Question 2, Rz is a left ideal of R . By Examples 3.2.1 (iv) zR is a right ideal of R . As $z \in Z(R)$, $Rz = zR$ so Rz is a 2-sided ideal of R . It is non-trivial as $z \neq 0$ and $z \cdot 1_R = z \in Z(R)$. Thus, as R is simple, $Rz = R$. Hence for some $r \in R$ we have $rz = zr = 1_R$. This r is a multiplicative inverse of z in R .

4. (i) $0_R \in I \cap J$, so $I \cap J$ is a non-empty subset of R .

Let $x, y \in I \cap J$, so $x, y \in I$ and $x, y \in J$.

Then $x - y \in I$ and $x - y \in J$, so $x - y \in I \cap J$.

Let $x \in I \cap J, r \in R$. Then $rx, xr \in I$ as I is an ideal and $rx, xr \in J$ as J is an ideal, so $rx, xr \in I \cap J$. Hence $I \cap J$ is an ideal of R .

(ii) $0_R = 0_I + 0_J \in I + J$, so $I + J$ is a non-empty subset of R .

Let $x, y \in I + J$, so $x = a_1 + b_1, y = a_2 + b_2$ for some $a_1, a_2 \in I, b_1, b_2 \in J$.

Then $x - y = (a_1 - a_2) + (b_1 - b_2)$. Now $(a_1 - a_2) \in I$ as I is an ideal, and $(b_1 - b_2) \in J$ as J is an ideal. Thus $x - y \in I + J$.

Let $r \in R, x = a_1 + b_1 \in I + J$.

Then $rx = ra_1 + rb_1$. Now $ra_1 \in I$ as I is an ideal and $rb_1 \in J$ as J is an ideal, so $rx \in I + J$. Similarly $xr \in I + J$. Hence $I + J$ is an ideal of R .

5. $A^2 = A$, so $A^2 - A = O$. Thus $A(A - I) = O$.

If A is non-singular, $A^{-1}A(A - I) = O$ so $A = I$.

If $A - I$ is non-singular, $A(A - I)(A - I)^{-1} = O$ so $A = O$.

Hence if $A \neq O$ and $A \neq I$ then A and $A - I$ are both singular.

6. (a) $(e + (1 - e)re)^2 = e^2 + e(1 - e)re + (1 - e)re^2 + (1 - e)re(1 - e)re$
 $= e + 0 + (1 - e)re + 0$ (since $e(1 - e) = e - e^2 = e - e = 0$)
 $= e + (1 - e)re$, so $e + (1 - e)re$ is idempotent.
 (b) $(1 - 2e)(1 - 2e) = 1 - 2e - 2e + 4e^2 = 1 - 4e + 4e = 1$, so $1 - 2e$ is self-inverse, hence it is a unit.

7. Let $a, b \in eRe$, so $a = ere$, $b = ese$ for some $r, s \in R$.

Then $a - b = e(r - s)e \in eRe$ and $ab = ere^2se \in eRe$, so eRe is a subring of R .

$ea = e^2re = ere = a$ and $ae = ere^2 = ere = a$, so e is the multiplicative identity in eRe .

Let $X = \{a \in R : ea = ae = a\}$. If $a \in X$ then $a = eae \in eRe$, so $X \subset eRe$.

Conversely, if $a \in eRe$ then $a = ere$ for some $r \in R$, so $ae = ere^2 = ere = a$ and similarly $ea = a$, so $a \in X$. Thus $eRe \subset X$, so $eRe = X$.

Taking $e = E_{11}$, we have $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} a & 0 \\ 0 & 0 \end{pmatrix}$,

so $eRe = \{aE_{11} : a \in \mathbb{R}\}$.

8. $9 \times 9 = 81 \equiv 9 \pmod{18}$, so $9 \times_{18} 9 = 9$, i.e. 9 is idempotent in $\mathbb{Z}_{18}$.

Thus $\mathbb{Z}_{18} = 9\mathbb{Z}_{18} \oplus (1 - 9)\mathbb{Z}_{18} = 9\mathbb{Z}_{18} \oplus 10\mathbb{Z}_{18}$,

$= \{0, 9\} \oplus \{0, 2, 4, 6, 8, 10, 12, 14, 16\}$.

$3 = 9 +_{18} 12$, and $9 \times_{18} 12 = 0$.